

User Acceptance Testing (UAT)

Project Title

Virtual Agent–Driven SLA Breach Awareness & Justification System

Introduction

User Acceptance Testing (UAT) is the final phase of testing conducted to verify that the **Virtual Agent–Driven SLA Breach Awareness & Justification System** meets the functional requirements and satisfies user expectations. The testing was performed from the perspectives of Support Agents, Team Leads, and Administrators to ensure that all workflows operate correctly and provide a reliable and user-friendly experience.

The primary objective of UAT was to validate the complete incident lifecycle, including SLA monitoring, risk classification, Virtual Agent interaction, AI-generated justifications, workflow automation, and reporting.

Objectives

The objectives of User Acceptance Testing were to:

- Verify that all functional requirements are implemented successfully.
 - Ensure users can perform their assigned tasks efficiently.
 - Validate the complete SLA monitoring workflow.
 - Confirm the accuracy of AI-generated SLA breach justifications.
 - Verify workflow automation and notifications.
 - Ensure dashboards and reports display accurate information.
 - Confirm that the application is ready for deployment.
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Test Environment

Parameter	Configuration
Platform	ServiceNow & Flask Application
Backend	Python (Flask)
Database	SQLite
Browser	Google Chrome
Operating System	Windows 11
AI Service	Google Gemini AI

Parameter	Configuration
Test Users	Support Agent, Team Lead, Administrator
Network	Stable Internet Connection

UAT Test Cases

Test Case 1 – User Login

Item	Description
Objective	Verify secure user authentication.
Expected Result	User successfully logs into the application.
Actual Result	Login completed successfully.
Status	Passed

Test Case 2 – Incident Creation

Item	Description
Objective	Verify incident creation functionality.
Expected Result	Incident is created and stored successfully.
Actual Result	Incident created successfully.
Status	Passed

Test Case 3 – SLA Monitoring

Item	Description
Objective	Verify continuous SLA monitoring.
Expected Result	SLA timer updates correctly in real time.
Actual Result	SLA monitored successfully.
Status	Passed

Test Case 4 – SLA Risk Classification

Item	Description
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Objective	Verify automatic SLA classification.
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Expected Result	Incidents are classified as SAFE, AT RISK, or BREACHED.
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Actual Result	Classification performed successfully.
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Status	Passed
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Test Case 5 – Virtual Agent Interaction

Item	Description
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Objective	Verify Virtual Agent assistance.
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Expected Result	Virtual Agent provides SLA information and guidance.
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Actual Result	Virtual Agent responded successfully.
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Status	Passed
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Test Case 6 – AI Justification

Item	Description
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Objective	Verify AI-generated SLA breach explanation.
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Expected Result	Google Gemini AI generates meaningful justification and recommendations.
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Actual Result	AI response generated successfully.
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Status	Passed
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Test Case 7 – Workflow Automation

Item	Description
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Objective	Verify automated Flow Designer workflows.
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Expected Result	SLA notifications and workflow actions execute automatically.
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Actual Result	Workflow executed successfully.
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Status	Passed
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Test Case 8 – Dashboard & Reporting

Item	Description
Objective	Verify dashboard updates and report generation.
Expected Result	Dashboards display accurate SLA statistics and reports.
Actual Result	Dashboard and reports generated successfully.
Status	Passed

UAT Summary

Module	Result
User Authentication	Passed
Incident Management	Passed
SLA Monitoring	Passed
SLA Risk Classification	Passed
Virtual Agent	Passed
Google Gemini AI Integration	Passed
Workflow Automation	Passed
Dashboard & Reporting	Passed

User Feedback

Support Agents

- The application is easy to use.
 - SLA notifications help prioritize incidents.
 - AI-generated guidance improves decision-making.
 - Virtual Agent interactions are helpful and intuitive.
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Team Leads / Managers

- Dashboards provide excellent visibility into SLA performance.
- Reports simplify monitoring and decision-making.
- SLA risk classification helps identify critical incidents quickly.
- Workflow automation reduces manual supervision.

Administrators

- User management is straightforward.
- Workflow configuration is efficient.
- Reports provide meaningful operational insights.
- Overall system performance is reliable and stable.

Acceptance Criteria

The application was accepted based on the following criteria:

- All functional requirements were successfully implemented.
- SLA monitoring operated correctly.
- AI-generated justifications were meaningful and accurate.
- Virtual Agent interactions functioned as expected.
- Workflow automation executed successfully.
- Dashboard information was accurate.
- No critical defects were identified during testing.

Observations

- User authentication operated successfully.
- Incident management workflows executed without errors.
- SLA monitoring remained accurate throughout testing.
- AI-generated recommendations enhanced user decision-making.
- Workflow automation reduced manual effort.
- Dashboards reflected real-time operational data.
- The application remained stable during testing.

Conclusion

User Acceptance Testing confirmed that the **Virtual Agent–Driven SLA Breach Awareness & Justification System** successfully satisfies user requirements and project objectives. The application effectively supports intelligent SLA monitoring, AI-assisted breach justification, workflow automation, and real-time reporting while providing a secure and user-friendly experience. Based on the successful completion of all planned test scenarios, the system is considered ready for deployment and demonstration.

